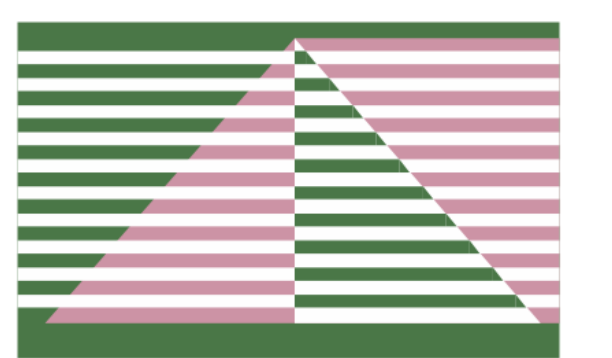


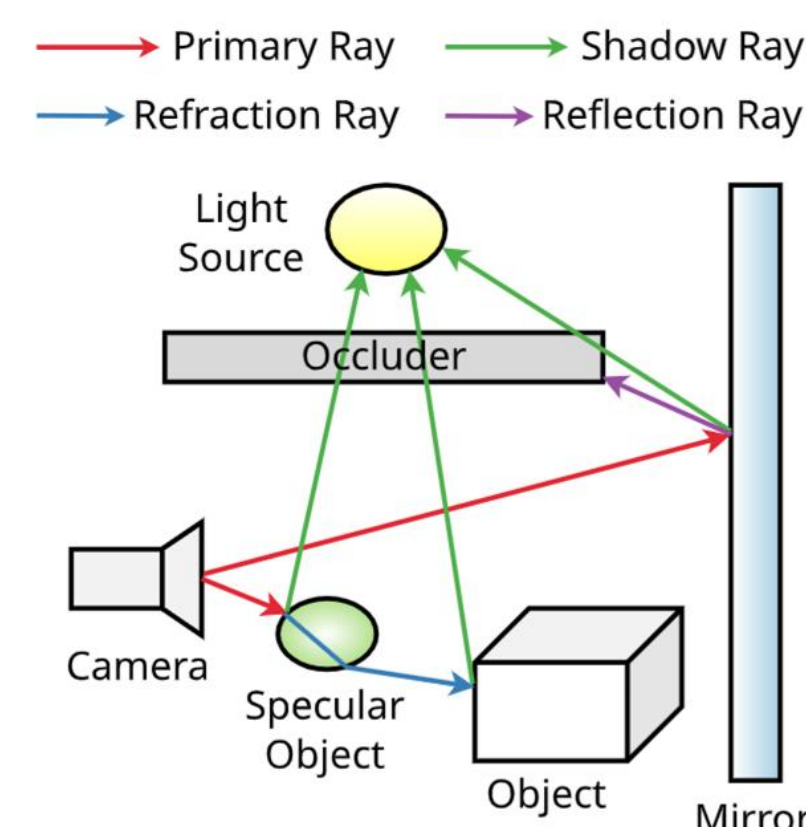
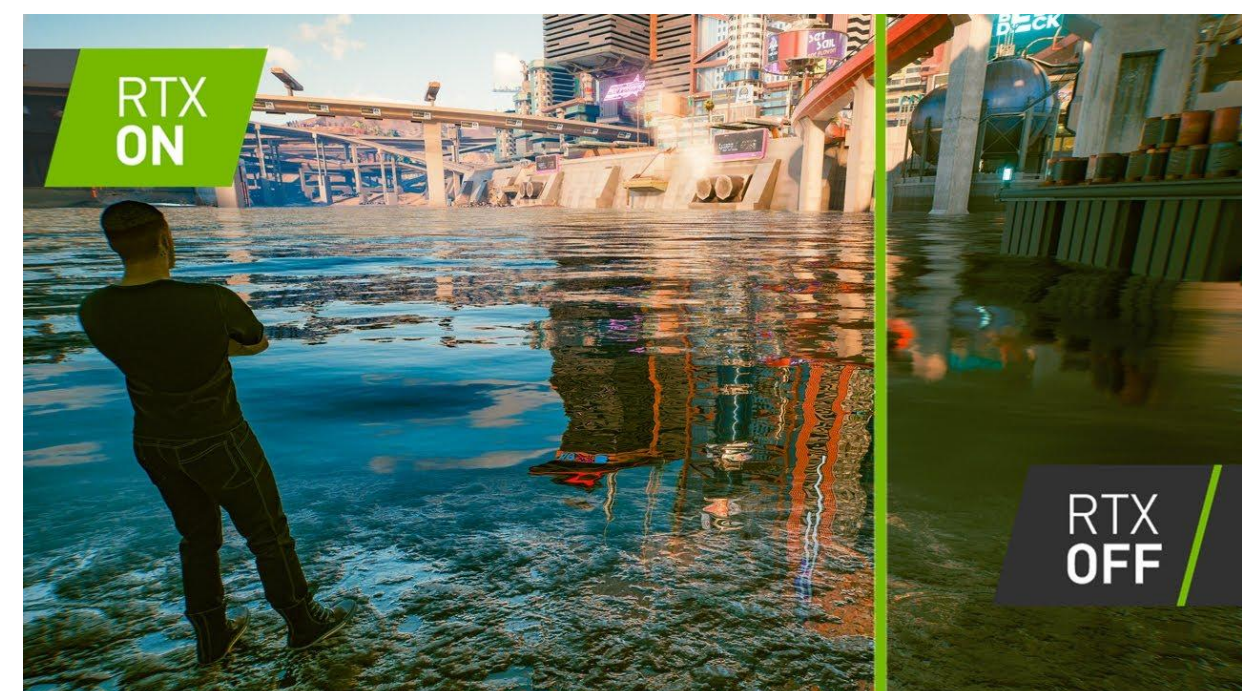


AQB8: Energy-Efficient Ray Tracing Accelerator through Multi-Level Quantization

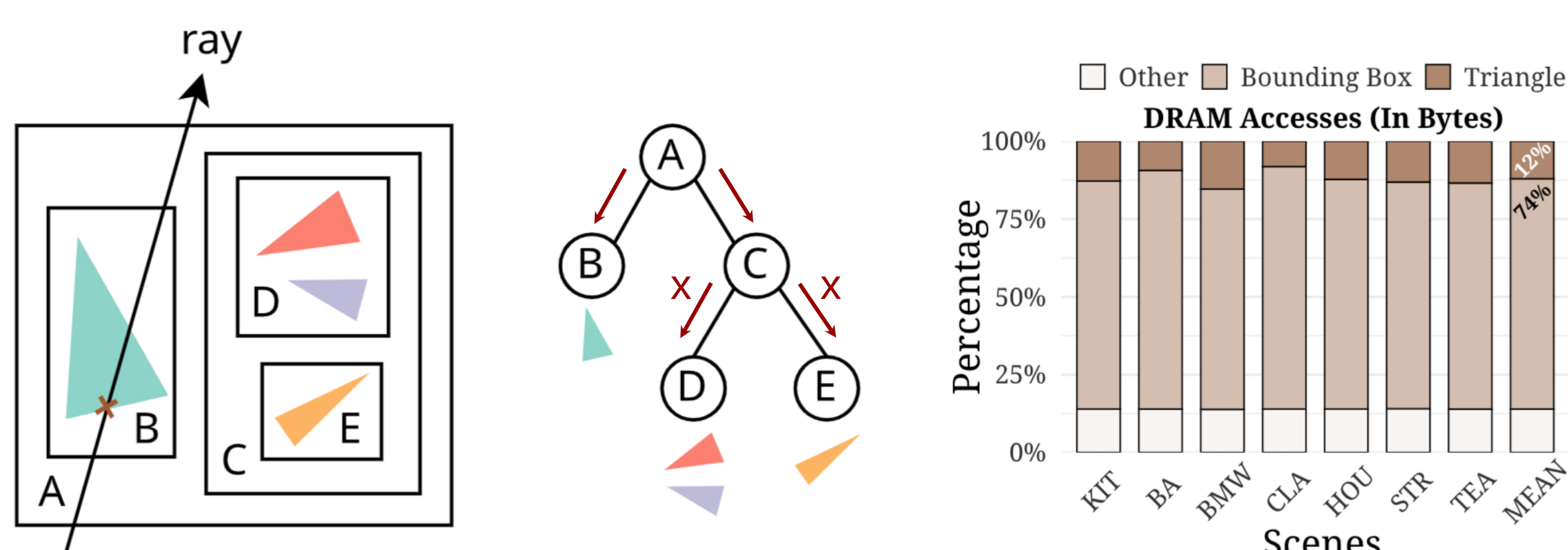
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Introduction & Challenge of Ray Tracing

- Ray Tracing (RT) is a graphics rendering method that **creates photorealistic images** by **simulating the physical path of light**

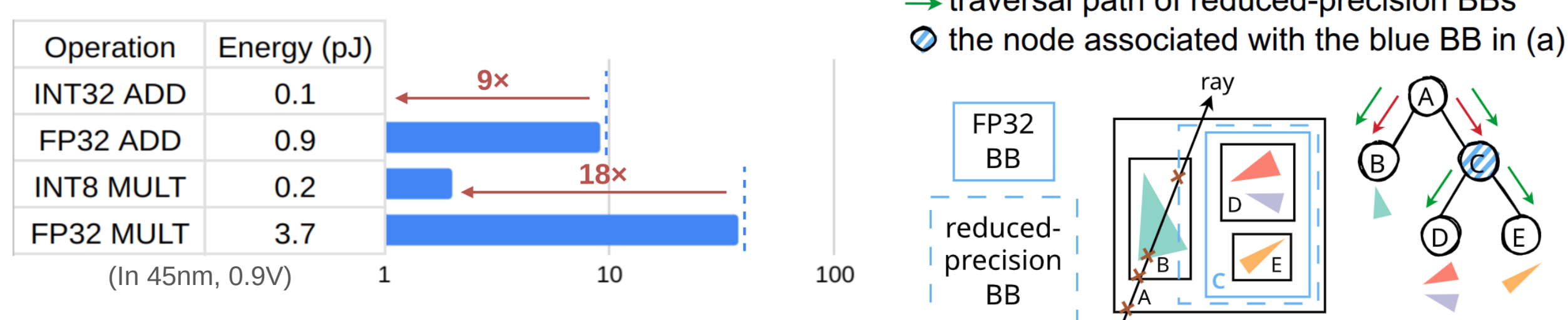


- RT uses a **Bounding Volume Hierarchy (BVH) tree** grouping objects into nested bounding boxes (BBs) to accelerate RT
- Challenges of RT:** The ray-BB intersection test often encode BBs as the **FP32** format, which consumes significant **memory bandwidth** (~74%) and **energy** of RT accelerator
- Goal:** Improving the energy efficiency of the RT accelerator by **quantizing BBs** and using **INT-based units** in RT accelerators



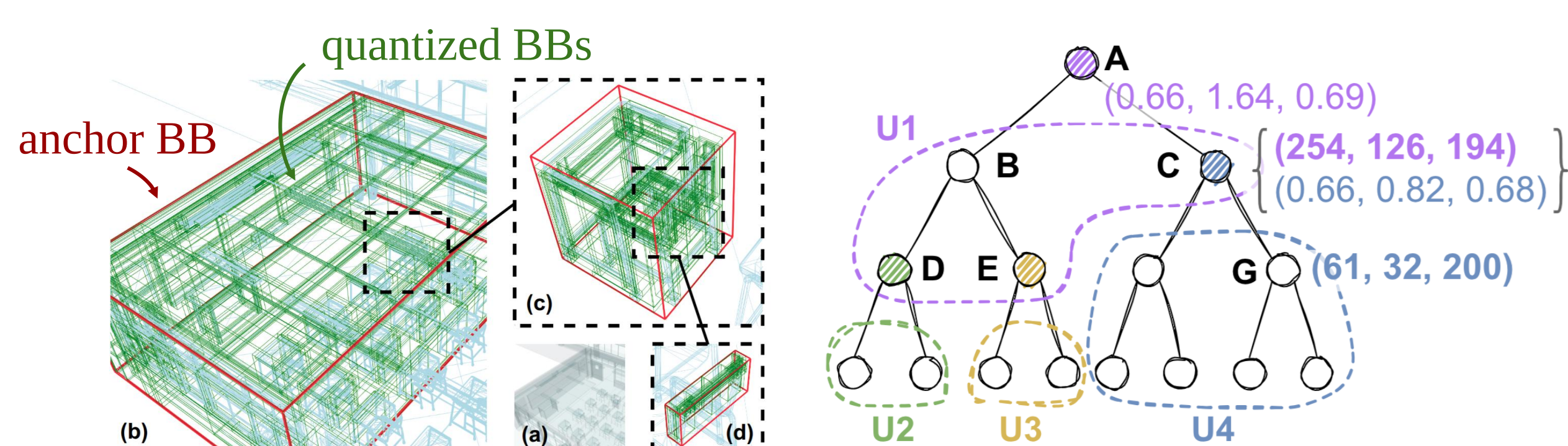
Why Existing Solutions Fall Short

- Compression Isn't Enough:** Prior work compresses BBs to low-bit formats (e.g., 8-bit) to reduce memory traffic. Must **decompress BBs back to FP32** for intersection tests, so the high energy cost of FP32 computations remains
- Quantization Error:** Directly encode BBs as low-bit formats (e.g., FP16) in the ray-BB intersection test will increase RT processing time



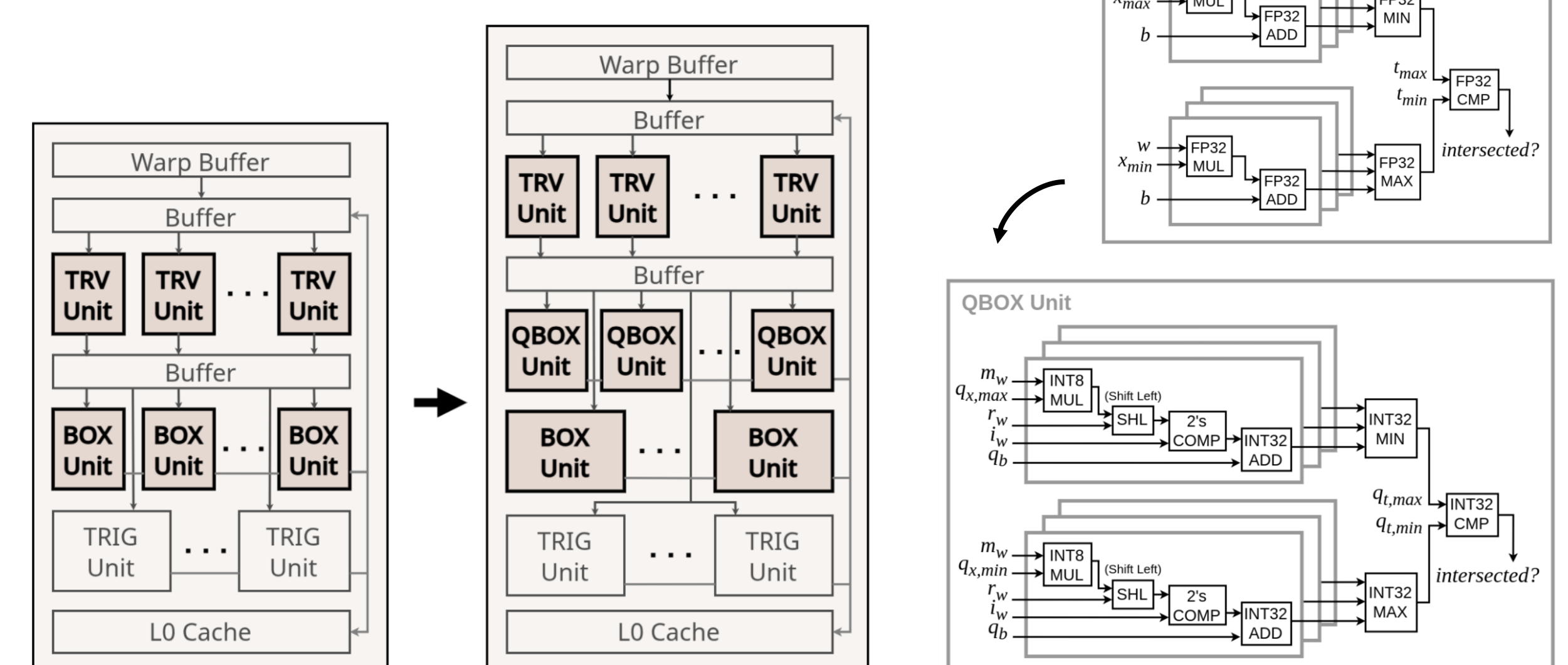
Our Solution: Multi-Level Quantization

- We introduce a mix-precision BVH tree structure that enables use **integer arithmetic on the ray-BB intersection tests without damaging graphics rendering results**
- Key Idea: Quantize BB locally within "clusters"**
 - Clusters** - Each cluster includes anchor and quantized BBs
 - Anchor BBs** - Coordinates are encoded as FP32 format
 - Quantized BBs** - Coordinates are encoded as INT8 format that is relative to their anchor BB



Our Proposed AQB8 Accelerator

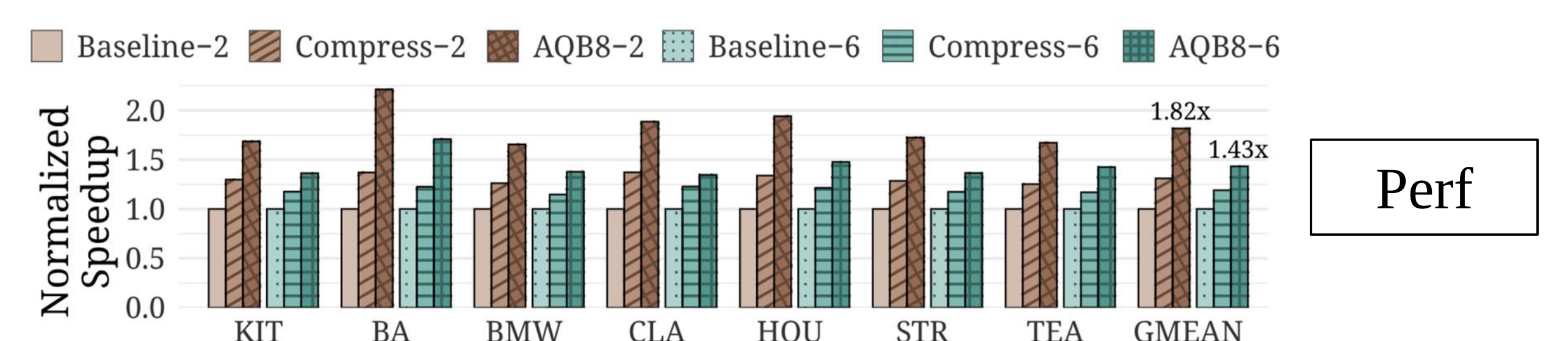
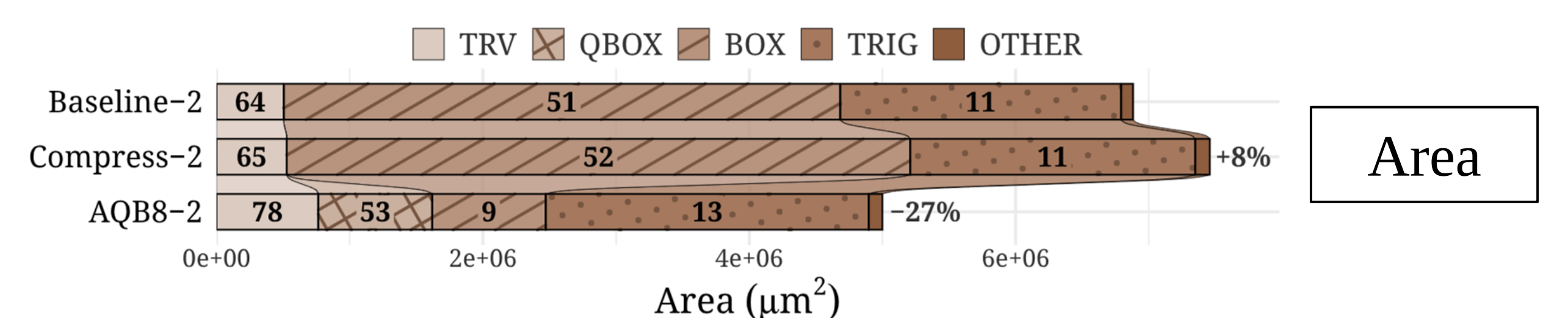
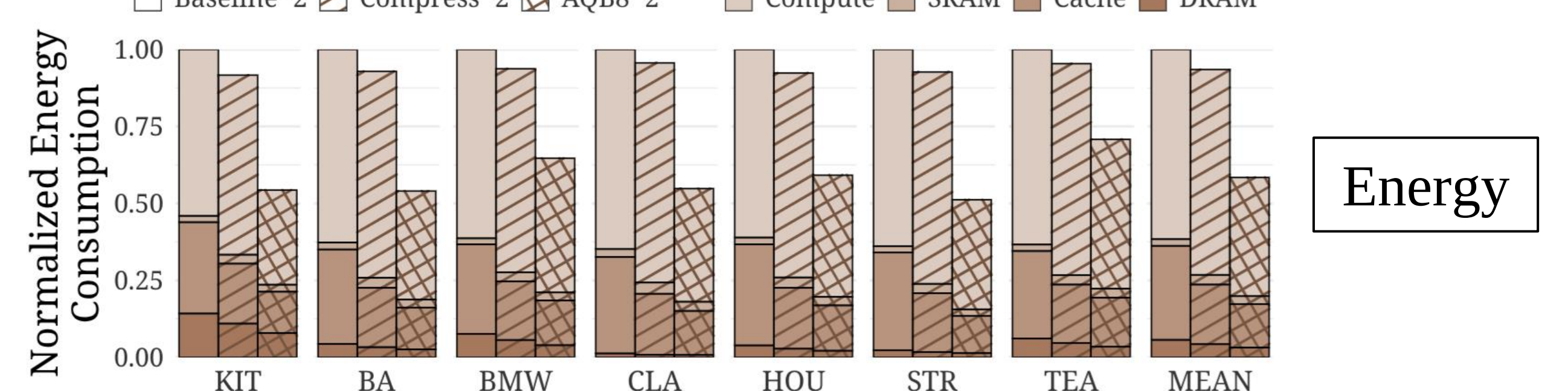
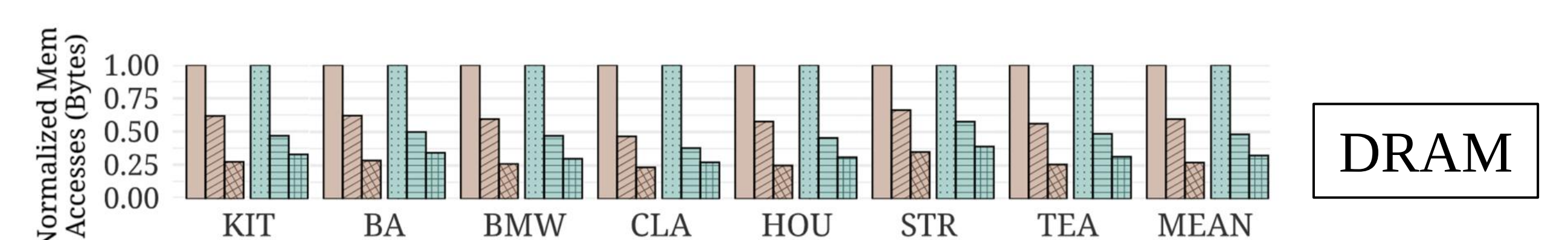
- We designed a RT domain-specific accelerator called AQB8, to improve the energy efficiency of RT processing
 - Original BOX Unit:** Conducts FP32 arithmetic for ray-BB intersections
 - New QBOX Unit:** We replace most BOX Units with our lightweight QBOX Unit, which performs INT8 arithmetic on quantized BBs using simple integer multipliers, shifters, and adders



- Memory Layout of BB:** AQB8 quantized node only takes **16 bytes** size instead of the **56 bytes** used in a standard BVH

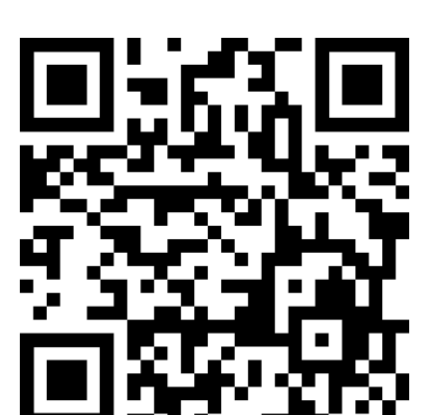
Key Results: Cheaper, Smaller, and Faster RT

- Compared to modern GPU RT accelerators, we achieve
 - ~70% Reduction in DRAM Accesses
 - ~49% Less Energy Consumption
 - ~27% Smaller Hardware Area
 - ~1.82x Performance Speedup



Conclusion

- This work presents a AQB8 RT accelerator that utilizes integer arithmetic on quantized BBs, reducing memory, energy, and hardware area while boosting performance of ray tracing through our multi-level quantization



Our Paper & Code!